INCH-POUND
MIL-DTL-32758
17 November 2023

DETAIL SPECIFICATION

BOOT, COMBAT, TROPICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tropical combat boot for use in both hot weather wet and dry environments.

1.2 Classification. The boot is available in one type and one class in the following sizes and widths, as specified (see 6.2).

1.2.1 Type.

Type I - Tropical combat boot

1.2.2 Class.

Class 1 - Coyote 498

1.2.3 Sizes and widths. The boots are sized using men's boot sizes.

Size: 3 through 16 (whole and half sizes)

Widths: Narrow (B) Regular (D) Wide (EE) X-Wide (EEEE)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52071 - Tape, Textile, Cotton, General Purpose

A-A-55093 - Laces, Footwear

A-A-55296 - Cloth, Twill, Cotton, 10.0 Ounces

A-A-59826 - Thread, Nylon

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8430

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-3122	-	Leather, Cattlehide, for Footwear Uppers and Gusset, Chrome Tanned, Fatliquored
MIL-DTL-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-PRF-17337	-	Webbing, Textile, Woven, Nylon
MIL-DTL-32075	-	Label: For Clothing, Equipage, and Tentage, (General Use)
MIL-DTL-32439	-	Cloth, Duck, Textured Nylon

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR - Protection of Environment

(Copies of this document are available online at www.ecfr.gov.)

U.S. ARMY DEVELOPMENTAL COMMAND - SOLDIER CENTER DRAWINGS

2-1-1635 - Speed Lace Assembly

(Copies of this drawing are available from the U.S. Army, DEVCOM, Soldier Center, ATTN: FCDD-SCD-PEI, 10 General Greene Avenue, Natick, MA 01760-5019.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Visual Assessment of Color Difference of Textiles

(Copies of this document are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers–Tension
ASTM D471	-	Standard Test Method for Rubber Property–Effect of Liquids
ASTM D624	-	Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers
ASTM D1052	-	Standard Test Method for Measuring Rubber Deterioration–Cut Growth Using Ross Flexing Apparatus
ASTM D1424	-	Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus

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ASTM D1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D2240	-	Standard Test Method for Rubber Property–Durometer Hardness
ASTM D3574	-	Standard Test Methods for Flexible Cellular Materials–Slab, Bonded, and Molded Urethane Foams
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3787	-	Standard Test Method for Bursting Strength of Textiles–Constant-Rate-of-Traverse (CRT) Ball Burst Test
ASTM D4786	-	Standard Test Method for Stitch Tear Strength, Single Hole
ASTM D4964	-	Standard Test for Tension and Elongation of Elastic Fabrics (Constant-Rate-of-Extension Type Tensile Testing Machine)
ASTM D4966	-	Standard Test Method for Abrasion Resistance of Textile Fabrics (Martindale Abrasion Tester Method)
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5963	-	Standard Test Method for Rubber Property–Abrasion Resistance (Rotary Drum Abrader)
ASTM D6797	-	Standard Test Method for Bursting Strength of Fabrics Constant-Rate-of-Extension (CRE) Ball Burst Test

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 17235	-	Leather – Physical and Mechanical Tests – Determination of Softness

(Copies of this document are available online at www.iso.org/home.html.)

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT (OECD)

OECD Guidelines for the Testing of Chemicals, Section 4, Test No. 404	-	Acute Dermal Irritation/Corrosion
OECD Guidelines for the Testing of Chemicals, Section 4, Test No. 406	-	Skin Sensitisation

(Copies of these documents are available online at www.oecd-ilibrary.org.)

SATRA TECHNOLOGY

SATRA TM1	-	Thickness of Leather and Insole Materials
SATRA TM64	-	Compression Set - Constant Stress Method
SATRA TM83	-	Measurement of the Area Shape Retention and Collapsing Load of Formed Box Toe (Toe Puff) and Counter (Stiffener) Materials
SATRA TM223	-	Floor Marking by Solings or Top Pieces

(Copies of these documents are available online at www.satra.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Standard sample. The finished leather shall be comparable to (see 6.5.2) a swatch of cattlehide leather for shade and sueded flesh appearance. Unless otherwise specified in this specification (see 3.5 and 3.14), the finished boot materials shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design. The boot is Coyote 498 in color and has a collar covered with a nylon cloth. The boot has a partial speed lace system consisting of eyelets, holes, or webbing loops below the comfort notch with the remainder being speed loops. The boot is lasted with a textile insole and contains a removable cushion insert. The exterior leather areas are flesh-side out leather with perforations for drainage/ventilation (see 3.5.8) located in the upper of each boot. The quarter is nylon duck; the backstay, pull tab, and ankle reinforcements are nylon webbing.

3.4.1 Expected service life. The expected service life of the boot is 12 months of normal wear. Extended wear on asphalt or other abrasive surfaces may reduce service life.

3.5 Materials.

3.5.1 Upper leather. With the exceptions specified in 3.5.1.1, the leather shall be flesh out drum dyed Coyote 498 cattle hide with color penetrating throughout the thickness of the hide so that the chrome tannage is not visible. The leather softness shall be 2.5 millimeters to 3.5 millimeters using a 35-millimeter aperture and shall meet requirements when tested as specified in 4.4.3. The leather for the vamps, outside counter pockets, and inside eyelet stays shall conform to MIL-PRF-3122, type I, class 4, treatment A and treatment B. The leather for the outside eyelet stays shall conform to MIL-PRF-3122, type II, treatment A.

3.5.1.1 Upper leather exceptions. The following exceptions to MIL-PRF-3122 shall be applicable for the upper leathers:

a. The flesh surface shall be suede.

b. The grain surface shall be lightly buffed to remove the surface of the grain only.

c. The finish requirements for the leather shall not apply.

d. The tear strength requirements for the inside and outside eyelet stays shall not apply.

e. The thickness for the cut parts shall be as specified in 3.14.1.1.

3.5.2 Insole. The Strobel lasting board shall be made with a non-woven water-repellent polyester material. The board shall meet the requirements of table I when tested as specified in 4.4.3.

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TABLE I. Strobel board performance requirements.

Characteristics	Physical Requirements
Weight, ounces per square yard (±10%)	15.5
Thickness, millimeters (minimum)	1.5
Tensile strength, psi (minimum)	
Machine direction	180
Cross-machine direction	180
Bursting strength, psi (minimum)	400
Elongation, percent (minimum)	
Machine direction	20
Cross-machine direction	10

3.5.3 Counters. The thermoplastic counter shall conform to the performance requirements of table II when tested as specified in 4.4.3. The counter shall fit the last used and shall fill the counter pocket area.

TABLE II. Thermoplastic performance requirements.

Characteristics	Physical Requirements
Initial collapsing load (N)	>130
Resilience, percent (minimum)	25
Moisture resistance, percent (minimum)	80
Area shape retention, percent (minimum)	
Initial	85
10th collapse	60
Thickness (inches)	0.068±10%

3.5.4 Box toe. A thermoplastic box toe shall be used and shall conform to the performance requirements of table II. The box toes shall be skived 3/8±1/8 inch wide at the breast. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.5.5 Upper construction. The upper shall be constructed with the materials and features below.

3.5.5.1 Cloth. The cloth for the quarters of the boot shall conform to MIL-DTL-32439, type III, class 1, style A. The air permeability requirement shall not apply for the cloth for both boots. The material shall be fungal resistant, water resistant, non-wicking, and backed with lining material to eliminate water absorption and moisture wicking. The color shall be Coyote 498 when compared to the standard sample (see 3.2). The cloth for the quarters may be treated with a non-fray finish to facilitate cutting and handling. The cloth for the collar cover may be flame-combined to a 0.05±0.008-inch thick layer of polyester foam having a density of 1.70±0.05 pounds per cubic foot. Inspection shall be in accordance with 4.4.2 and testing shall be as specified in 4.4.3. Narrow goods, including back strap, ankle reinforcements, and binding tape, may be the same nylon fabric construction as that used in the gusset and quarter side panels. Nylon, as defined in the applicable type, class, and style, may be used in the tongue area to improve durability and comfort as long as there is no degradation in performance.

3.5.5.2 Gusset fabric. The fabric for the gusset shall be a 100-percent polyester spacer knit. The face fabric shall be a jersey knit, and the back fabric shall be mesh. The mesh shall be a raschel warp-knit mesh with fillet stitch. The color shall be Coyote 498 when compared to the standard sample (see 3.2). The gusset fabric may be treated with water repellency. The gusset fabric shall conform to table III. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

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TABLE III. Gusset fabric performance requirements.

Characteristics	Physical Requirements
Weight, ounce per square yard	12.5±10%
Thickness, inches	0.116±10%
Wales (width)	25±3
Courses (length)	52±5
Breaking strength (pounds) (minimum)	
Wales (width of fabric)	75
Courses (length of fabric)	75
Tear strength (pounds) (minimum)	
Wales (width of fabric)	10
Courses (length of fabric)	8
Elongation (percent) (minimum)	
Wales (width of fabric)	125
Courses (length of fabric)	90

3.5.5.3 Fabric liner for cushioned insert and inside counter pocket. The cloth for the fabric top cover on the cushioned inserts and the inside counter pocket shall conform to the requirements in table IV. Inspection shall be in accordance with 4.4.2 and testing shall be as specified in 4.4.3.

TABLE IV. Fabric liner performance requirements.

Characteristics	Physical Requirements
Martindale abrasion, dry (153,600 cycles) (minimum)	9,600
Bursting strength, pounds (minimum)	125

3.5.5.4 Back seam tape. The tape shall conform to A-A-52071, type I, class 3, 1/2±1/32 inch wide. As an alternate, the tape shall conform to MIL-PRF-5038, type III, class 1, 1/2±1/32 inch wide. Back seam tape shall not extend over the comfort collar. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.5.5.5 Vamp lining. The cloth for the vamp lining and optional inside counter pocket shall be a textile knit laminate conforming to the requirements in table IV. As an alternative vamp lining, cotton twill conforming to class 2 of A-A-55296 with an added durable water repellent finish may be used, so long as the end item meets the requirements stated in table IV. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.5.5.6 Binding tape. The tape for binding the top edge of the gusset shall be nylon conforming to MIL-PRF-5038 type III, class 1 or 2, 3/4 inch. The color shall be Coyote 498 matching (see 6.5.2) that of the standard sample (see 3.2).

3.5.5.7 Backstay and ankle reinforcement tape. The backstay tape shall be cut from 1-inch wide nylon webbing. The ankle reinforcing tape width shall be between 2 and 3 inches of nylon webbing, starting at the corner of the heel counterpart and back seam tape, crossing over the ankle bone when worn. The webbing shall conform to MIL-PRF-17337, class 1 or class 2. The color shall be Coyote 498 (see 6.5.2) when compared to the standard sample (see 3.2).

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3.5.5.8 Pull tab. The boot shall contain a pull tab loop. The nylon tape at the back seam shall be extended to create a pull tab with a circumference of 3 to 3-3/4 inches when completed. The stitching of the pull tab shall be below the padding of the comfort collar. The top of the pull tab shall be even with the top of the collar (+0, -1/4 inch). The webbing shall be tested in accordance with 4.4.3.

3.5.5.9 Comfort collar. The foam rubber for the padded collar shall be 1/4-inch thick latex foam having a density of 6.7±2 pounds per cubic foot. One side of foam shall be covered with a cotton scrim cloth.

3.5.5.10 Comfort notch. The comfort notch shall be a V-shaped cut out between the eyelets and the speed loops that allows for both plantar- and dorsi-flexion of the foot.

3.5.6 Shank. The boot shall include a fiberglass shank bonded to the insole, outsole, or both. The fiberglass shank (glass and resin area), when cured, shall be 3/4±1/4 inch in width. The rear end of the shank shall be placed 2-1/4±1/4 inches in front of the rear end of the boot, the forward end of the shank shall not extend into the ball area, and the shank shall be oriented along the center line of the boot. The shank lengths shall conform to the schedule as specified in table V.

TABLE V. Shank lengths.

Last Size (All Widths)	Shank Length (Inches)	
	Minimum	Maximum
Last sizes 3 through 4.5	2-3/4	3-1/8
Last sizes 5 through 8	3-1/2	4
Last sizes 8.5 through 11.5	4-1/8	4-3/8
Last sizes 12 through 13.5	4-1/2	4-3/4
Last sizes 14 through 16	4-7/8	5-1/4

3.5.7 Eyelets and speed lacing loops. The size AA eyelets, and the speed lacing loops produced in accordance with Drawing 2-1-1635, shall be brass or aluminum with a durable coated finish in Coyote 498 (see 6.5.2), which is durable enough to be expected to remain on the eyelets and speed laces for the life of the item (see 3.4.1). The eyelets and speed laces shall be expected to remain permanently fixed to the upper, and shall not rust, tarnish, or otherwise change the subdued finish color for the life of the item when exposed to standard operational environments in tropical climates. Adjustments shall be made such that all sizes allow users to easily utilize the speed lace system and proper closure of the item is achieved. The number of speed laces and eyelets shall be in accordance with table VI and shall be equally spaced on the eyelet stays to allow proper lacing and cinching of the boot for ease of donning and doffing, and shall have the same alignment between both rows. The bottom eyelets shall be placed in a position on the foot such that they do not cause hot spots, blisters, or other irritation. The distance between eyelets and speed loops shall be not greater than 1-3/8 inches, center to center, from one another. As an alternative, webbing for the lower lacing portion and at least three sets of speed lacing eyes for the upper lacing portion may be used.

TABLE VI. Speed loops and eyelets.

Boot Sizes	Below Comfort Notch	Above Comfort Notch
	Number of Eyelets	Number of Speed Loops
3 to 12	3	3 to 4
12.5 to 16	3	4 to 5

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3.5.8 Drainage. The boot shall have a means of allowing drainage and ventilation but not allowing sand to get into the boot, on both the medial and lateral sides of the leather vamp, behind the ball area of the foot, and the natural flex line of the boot. Either perforated holes (see 3.5.8.1) or metal grommets with an imbedded screen (see 3.5.8.2) may be used. A combination of both methods may be used with the approval of the design by the contracting agency.

3.5.8.1 Drainage and ventilation holes. If perforations are used, there shall be no less than eight perforations on each side for sizes 3 through 7.5 and no less than ten perforations on each side for sizes 8 through 16. There shall be no more than fifteen perforations on each side in any size. The diameter of the perforations shall be not less than 0.04 inch and equally spaced from one another on each side using a design pattern of two to three rows and the necessary number of columns. The number of perforations in each row shall not differ from all other rows by more than one. The drainage perforations may be backed by a non-metallic screen mesh or knit laminate material with a sieve size of not greater than 0.032 inch (0.8 millimeter). The mesh or knit laminate, if used, shall be held in place with adhesive, stitching, or both around the perimeter of the perforations. Internal components of the boot, to include adhesives, shall not interfere with the proper drainage from the perforation holes. The lowest row of drainage perforations shall be no further than 0.375+0.125 inch from the midsole. Perforations, and any backing materials, shall not present hot spots or other sources of discomfort to the wearer, nor shall they cause any degradation to the durability of the boot. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.5.8.2 Drainage and ventilation grommets. Grommets, if used, shall be permanently fixed to the boots for the expected life of the item (see 3.4.1) and shall have a subdued finish to blend with the color of the leather upper. The grommets shall incorporate a permanently-embedded screen mesh with a sieve size not greater than 0.032 inch (0.8 millimeter). The placement of the grommets shall be in the inner arch of the boot at a location that allows the greatest possible drainage of water from the boot. Internal components of the boot shall not interfere with the proper drainage and ventilation through the eyelets and shall not create hot spots to the wearer. Eyelets shall be placed at the same location relative to the solid midsole on all boots (with variation according to boot size). Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.6 Laces. The laces shall conform to type III, class 1 of A-A-55093. The color of the laces shall be Coyote 498 when compared to the standard sample (see 3.2). The length of the laces shall be a minimum of 66 inches for all sizes. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.7 Soling system requirements.

3.7.1 Cup sole. If a cup-type soling system is used, the sole of the boot shall be a single unit cup-type outsole. The method of attachment shall be adhesive or sewing, or a combination of both, using industry standard methods. The cup-style outsole shall have a tread pattern that is self-cleaning while allowing for the incorporation of design improvements to improve traction and durability. The sole shall be attached with a cement construction, sole stitch construction, or both. The outsole shall incorporate a shank (see 3.5.6) to provide stiffness and strength to the outsole longitudinally. Outsoles may include a layer or material to provide additional protection against puncture. Although not lab tested, the outsole color should be similar to the color of the nylon fabric (see 3.5.5.7). See 6.7.2 for a sole design that has demonstrated the ability to meet the soling requirements.

3.7.1.1 Cup sole bonding. The outsole shall be bonded to the upper using stitching, adhesive, or a combination of stitching and adhesive. The bond shall be continuous with no gaps or open spaces and the outsole shall be bonded such that it cannot be separated from the upper by hand. The cup sole rubber compound shall meet the requirements of 3.7.3.

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3.7.2 Direct attach sole. The boots shall be made on military last model "FMT U3813-1 Military", or equal, designed for direct attach based on Government approval. When using the FMT U3813-1 model, the numerical size of the boot shall correspond with one more than the numerical size of the last used to produce the boot (i.e., a size 11R boot shall be produced on a size 10R last). The direct attach last has a common last bottom shared by N and R lasts and a common last bottom shared by W and XW lasts for each whole and half size. The direct attach outsole mold shall produce an outsole with a tread pattern that is self-cleaning while allowing for the incorporation of design improvements to improve traction and durability. The outsole shall meet the requirements of 3.7.3, and the cushion midsole shall meet the requirements of 3.7.4. See 6.7.2 for a sole design that has demonstrated the ability to meet the soling requirements.

3.7.3 Rubber outsole compound requirements. The requirements for the rubber outsole compounds are based on testing done on slabs and not actual soles. The requirements shall be as specified in table VII when tested in accordance with 4.4.3.

TABLE VII. Rubber outsole compound requirements.

Characteristics	Requirement
Abrasion (minimum)	150
Hardness (shore A)	70-80
Stitch tear, psi (minimum)	
Dry	180
Fuel B	50
Die C tear strength, pounds (minimum)	308
Ross flex, IRM 903, percent (maximum)	
50,000 flexes	250
Volume swell percent (maximum)	
Fuel A (24 hours)	12
Non-marking	Pass
Tensile, pounds (minimum)	2,100
Elongation, percent (minimum)	400

3.7.4 Cushioned midsole. The cushioned midsole used when incorporating a direct attach sole shall meet the requirements of table VIII.

TABLE VIII. Cushioned midsole physical requirements.

Characteristics	Minimum	Maximum
Density, grams per cubic centimeter	0.40	0.60
Hardness (shore A)	50	75
Tensile strength, kilogram-force per square centimeter	25	--
Elongation, percent	250	--
Tear strength, kilograms per centimeter		
Die T	3.7	--
Volume swell, percent		
w/IRM 903 (at 24 hours)	--	10
Compression set, percent	--	20
Ross flex, percent at cycles (1 inch x 6 inches) at -5 °C	--	900 at 150,000

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3.8 Thread, nylon. The thread shall be nylon, conforming to A-A-59826, class A or class B, type I, Tex size 68-70 (Government size E); type II, Tex size 70-76 (Government size E); or type III, Tex size 70-75 (Government size E). The color of the thread shall be Coyote 498. Colorfastness requirements shall not apply. Inspection shall be in accordance with 4.4.2, and testing shall be as specified in 4.4.3.

3.9 Removable cushion insert. The removable cushion insert for the boot shall be a low-density polyether-polyurethane heel cup support made of a polyurethane system, and lined (black or tan in color). The cushioned insert shall provide comfort and shall be made to optimize the fit of the item, using the last the item was made on. The cushion insert should take a minimal compression set, even after extensive wear or material conditioning. The cushion insert and its lining shall be resistant to fungal growth, shall provide good frictional characteristics, and shall not be susceptible to hydrolysis. Characteristics shall be equal to or better than the cushioned insert in currently fielded hot weather boots.

3.10 Vamp design. The vamp design shall allow for the leather vamp to extend up the gusset to approximately the second eyelet of the boot.

3.11 Boot height. The height of the boot shall be between 8.5 and 9.0 inches for a size 10R boot from the feather line to the top of the boot at its tallest point. The height shall be graded up and down between sizes and widths appropriately to minimize the interference with the calf musculature. The overall height of the boot shall include the collar.

3.12 Boot weight. When tested as specified in 4.4.4, the weight of the boot shall not exceed 1.8 pounds (dry), with an objective weight of 1.5 pounds, for a finished men's boot made on a men's size 10R last. The weight of the boot when saturated with water should be minimized through materials and construction technique. Materials that reduce wet or dry weight with any detrimental effect on durability, comfort, breathability, or any other aspect of overall performance are not in the best interest of the United States military and should not be used.

3.13 Lasts and molds. Last sizes 3 through 16 cover the boot size range of 3 through 16 in this specification.

3.13.1 Direct attach soling construction lasts and molds. The boots shall be lasted and molded on commercial lasts designed for direct attach. The numerical size of the boot shall correspond with one more than the numerical sizes, and in the appropriate width, of the last used to produce the boot. The direct attach process (either injection molding or open pouring) shall utilize a separate cushion midsole mold to produce appropriate widths for each whole and half size.

3.14 Construction.

3.14.1 Uppers.

3.14.1.1 Cutting leather uppers. The leather parts shall conform to the thickness requirements specified in table IX. All leather for boot components, with the exception of the inside eyelet stays, shall be cut so as to be used flesh side out. Vamps shall be examined as specified in 4.4.3. Outside eyelet stays shall not be split down parts. Skiving of leather upper parts is permitted.

TABLE IX. Thickness of leather uppers (ounces).

Characteristics	If Not Split Down		For Split Down Parts	
	Minimum	Maximum	Minimum	Maximum
Vamp	4	5	--	--
Outside counter pocket	4	5	--	--
Outside eyelet stays	3	4	--	--
Inside eyelet stays	3	4	3-1/2	4

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3.14.1.2 Cutting fabric upper parts. The quarters shall be cut in the bias direction such that the collar seam is not parallel with either the warp or fill direction. The collar cover shall be cut in the bias direction.

3.14.2 Insole construction. The boot shall be made with a Strobel construction in which the bottom construction of the boot shall include a non-woven textile insole that is stitched 360 degrees around the bottom of the upper. The textile insole should match the last bottom gauge.

3.14.3 Marking, permanent identification. With the exception of Product Demonstration Models, boots shall be permanently marked on the inside of the boot in the shaft or gusset area with the appropriate size and width, the contract number, and the company or brand name. Alternatively, the marking may be applied to the inside of the upper portion of the tongue of the boot. The marking may be in the form of a label that will last for the expected life of the boot, and it shall not reduce footwear functionality. The boot shall contain a manufacturing tracking lot number and a born date. Embossing of a company or brand name or application of a company branded label on the exterior of the boot shall not be permitted. Unit packs shall include the National Stock Number (NSN) and Universal Product Code barcodes as specified in the contract (see 6.2). Each symbol shall be human and scanner readable. As an alternate, a sewn in label may be utilized.

3.15 Assembly.

3.15.1 Boot finishing. No top finish or treatment shall be applied.

3.15.2 Pairing, lacing, and instruction tag. The boots shall be paired. Each pair of boots shall consist of a left and a right boot of the same size. Measurements shall not vary between boots of a pair by more than 1/4 inch for any measurement. The shade of all leather parts in a pair of boots shall be a close match. A lace shall be inserted through the top speed lace loop of the outside quarter of each boot, and the two laces for each pair shall be tied firmly together. One instruction tag shall be attached to each pair of boots by inserting a lace through the hole in the instruction tag. If not incorporated into the instruction tag, a barcode tag (see 3.16) shall also be attached to each finished pair of boots.

3.15.2.1 Instruction tag. An instruction tag shall be attached to each pair of finished boots. The tags shall be printed on tag stock. The tag shall be legible but shall not exceed 4-1/2 inches in any dimension, to include all pertinent instructions for the user. The tag may be folded or otherwise oriented to accommodate all necessary information. A punch hole shall be made in the tag for insertion of the lace. The barcode (see 3.16) may be incorporated into the instruction tag.

3.16 Barcode tag. The barcode tag shall conform to MIL-DTL-32075, type VIII, class 17. The barcoding element shall be a 13-digit NSN. The barcode type shall be a medium to high code density. The barcode tag shall be located so that it is completely visible on the item when it is packaged as specified and in a manner that causes no damage to the item. As an option, a hang tag identifying the features of the boot is permitted.

3.17 Toxicity. When specified (see 6.2), boots shall be tested for the presence of health hazards. The finished boot shall not present a health hazard and shall show compatibility with prolonged, direct skin contact, as specified in 40 CFR §798.2250, Dermal Toxicity, and 40 CFR §798.4100, Dermal Sensitization (see 4.7). Chemicals recognized by the Environmental Protection Agency as human carcinogens shall not be present in the finished boot.

3.18 Dry time. When tested in accordance with 4.5.1, the boots shall dry at a rate of 35 percent water removal in 6 hours, representing a 10-percent increase in water removal compared to the current combat boot (see 6.7.1).

3.19 Workmanship. The finished boots shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the contract (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.1.1 Inspection conditions. Unless otherwise specified in this specification or in applicable contract or procurements documents (see 6.2), all inspections shall be performed in accordance with the test conditions specified in this specification and reference documents.

4.2 First article inspection. The first article (see 3.1.1 and 6.3) shall be inspected for compliance with design, workmanship, and dimensions. In addition, the first article shall include verification of compliance to the visual inspection (see 4.4.2), component testing (see 4.4.3), end item testing (see 4.4.4), and shade (see 4.5). The presence of excessive defects in excess of the acceptance quality limit (AQL), as defined by contract or purchase order (see 6.2), or failure to pass any test shall be cause for rejection of the first article. The sample size shall be specified in the procurement document (see 6.2). The sample unit shall be one pair of boots and the lot size shall be expressed in pairs of boots.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4.1 and 4.4.2 and the tests of 4.4.3 and 4.4.4, as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with AQLs as specified in the contract or order (see 6.2).

4.4 Component, material, and end-item inspection.

4.4.1 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether the components are as specified or operations and assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2 End-item visual examination. The end items shall be examined for the defects listed in table X. The lot size shall be expressed in units of boots. The removable cushion inserts shall be inserted in the boots. For the pairing examination, and when determining possible differences in outsole thickness, the pair shall be examined together. During the inspection for the defects listed, the minor defects or departures from specification requirements listed in the table shall be observed and recorded. Presence of these departures shall not result in the rejection of the lot, but the plant quality assurance and the Government personnel reserve the right to establish and impose lot rejection criteria when either the frequency or severity of the presence of these departures warrant. These additional rejection criteria may be invoked at any time during the life of the contract when the Government judges this action to be in their best interest. Defects of pairing shall be classified as a single defect. Note: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more.

TABLE X. Defects.

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship (general) affixed	Any component missing or not specified type	101	
	Any component misplaced or not as specified		
	Seriously affecting serviceability	102	
	Not seriously affecting serviceability		201
	Vamp stitched more than 5/32 inch from edge at throat	103	
	Vamp stitched more than 3/32 inch from edge at throat but not greater than 5/32 inch from edge at throat		
	Seriously affecting serviceability	104	
	Not seriously affecting serviceability		202
	Vamp throat or wings not skived		
	Seriously affecting serviceability	105	
	Not seriously affecting serviceability		203

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TABLE X. Defects - Continued.

Examine	Defect	Examine	
		Major	Minor
Quality of leather	Thickness more than 1/2 ounce less than the minimum specified	106	
	Thickness less than the specified minimum but not exceeding 1/2 ounce less		204
	Thickness more than maximum specified (see 3.14.1.1)	107	
	Off-stretch cut		
	Seriously affecting serviceability	108	
	Not seriously affecting serviceability		205
	Slaughter cut		
	Seriously affecting serviceability	109	
	Not seriously affecting serviceability		206
Leather	Any spot or stain more than 1 inch in its longest dimension	110	
	Any hole that is not part of the boot design, cut, or tear	111	
	Missing end or pick, thick or thin areas		207
	Overall uncleanliness		208
	Crease or hard crease		209
	Slubs, knots, slugs, or gout		210
	Shade bar, streaks or uneven dying		211
	Not flesh side out or cattle hide nubuck	112	
	Leather deeply snuffed fiber structure damaged		212
	Grub or tick damage, scratches, brands, or bony leather	113	
	Fat wrinkles or veins		213
	Stretchy vamp		214
	Rough fiber on flesh side		215
	Excessive roughing or scouring		216
	Water repellent treatment not uniform		217
	Adhesive bleeding greater than 1/8 inch on leather		218

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TABLE X. Defects - Continued.

Examine	Defect	Examine	
		Major	Minor
Outsole	Excessive adhesive		219
	Defects in tread or edge area not otherwise specified		220
	Incomplete or incorrect bonding of outsole to upper (i.e., not securely attached)		
	Seriously affecting serviceability	114	
	Not seriously affecting serviceability		221
	Outsole not aligned with boot edges	115	
	Incorrect tread pattern	116	
	Tread lugs not properly formed		222
	Thickness not as specified		223
	Edge irregular, affecting appearance		224
	Any malformation		225
Insole	Not correct size for the boot		226
	Textile insole is not securely stitched to upper	117	
	Any protruding lasting stitch	118	
	Polyurethane bleeding through		
	Seriously affecting serviceability	119	
	Not seriously affecting serviceability		227
	Wrong foot/pairing issue		228
	Missing		229
Laces	Frayed		230
	Cut		231
	Ends not properly finished		232
Labels and marking	Label missing, incorrect, or illegible	120	
	Required information missing from the label	121	
	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size or not readable by scanner		233
	Causes damage to end item	122	
Barcode tag	Barcode omitted or not readable by scanner	123	
	Human-readable interpretation omitted or illegible		234
	Barcode type not as specified		235
	Code density not as specified		236
Packaging	Not packaged in accordance with section 5 of this specification		237

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TABLE X. Defects - Continued.

Examine	Defect	Examine	
		Major	Minor
Pairing	Different size boots	124	
	Two left or two right boots in a pair	125	
	Shade differences between boots		238
	Boots from different lots		239
	Dimensional differences between boots (except factors below):		240
	More than 1/4 inch	126	
	More than 1/8 inch but less than 1/4 inch		241
	Variation of more than 1/4 inch in height of pair	127	
	Difference in outsole thickness between left/right boot		
	More than 3/32 inch	128	
	More than 1/16 inch but less than 3/32 inch		242
Construction and workmanship	Any damage not otherwise specified affecting appearance		243
	Design not as specified	129	
	Color not as specified; finish streaky, chipped, flaky, or shaggy		244
	Nappy on upper		245
	Any raw edges not stained to match upper leather	130	
	Cut, tear, hole, repair, or factory damage		
	Seriously affecting serviceability	131	
	Not seriously affecting serviceability		246
	Wrinkled or bunched area at the back seam		
	Seriously affecting serviceability	132	
	Not seriously affecting serviceability		247
	Component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein)		
	Seriously affecting serviceability	133	
	Not seriously affecting serviceability		248
	Any component, assembly, or design present creating “hot spots”	134	
Bottom of boot	Any component missing, sized wrong, malformed, or not specified type	135	
	Poor heel seat, side, or toe lasting		
	Seriously affecting serviceability	136	
	Not seriously affecting serviceability		249
Upper part of boot	Upper damaged	137	
	Uppers not firmly pulled down to last		
	Seriously affecting serviceability	138	
	Not seriously affecting serviceability		250

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TABLE X. Defects - Continued.

Examine	Defect	Examine	
		Major	Minor
Gusset polyester - mesh spacer fabric	Mended or weak spot	139	
	Fabric oriented incorrectly or mesh-side out	140	
	Abrasion, spot, stain, or discoloration		
	Seriously affecting serviceability	141	
	Not seriously affecting serviceability		251
Binding	Omitted	142	
Nylon quarter	Mended or weak spot	143	
	Two or more yarns missing	144	
	Quarter not cut on the bias	145	
	Abrasion, spot, stain, or discoloration		
	Seriously affecting serviceability	146	
	Not seriously affecting serviceability		252
Webbing (ankle reinforcement)	Not firmly and tightly woven, edges frayed or scalloped	147	
	Not properly placed	148	
	Incorrect width	149	
Seams and stitching (upper)	Open seam not repaired		
	Seriously affecting serviceability	150	
	Not seriously affecting serviceability		253
	Quarter panel and gusset fabric not caught in second row of inside eyelet stay stitching		254
	One or more run-off stitches	151	
	Loose tension resulting in a loosely secured seam		
	Seriously affecting serviceability	152	
	Not seriously affecting serviceability		255
	Tight tension resulting in puckering or cutting of leather		
	Seriously affecting serviceability	153	
	Not seriously affecting serviceability		256
	Row of stitching with less than eight stitches per inch but not less than six stitches per inch		257
	Any row of stitching with less than six stitches per inch	154	
	More than 10 stitches per inch		
	Seriously affecting serviceability	155	
	Not seriously affecting serviceability		258
	Gauge of stitching not as specified or irregular		
	Seriously affecting serviceability	156	
	Not seriously affecting serviceability		259

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TABLE X. Defects - Continued.

Examine	Defect	Examine	
		Major	Minor
Seams and stitching (upper) (continued)	Stitching omitted where required		
	Seriously affecting serviceability	157	
	Not seriously affecting serviceability		260
	Thread ends not trimmed throughout boot		
	Seriously affecting serviceability	158	
	Not seriously affecting serviceability		261
	Needle holes or needle chew		
	Seriously affecting serviceability	159	
	Not seriously affecting serviceability		262
	Sand hole (i.e., tongue not properly caught in at throat)		263
Vamp or counter pocket stitching at outsole junction	One or more rows of defective stitching not repaired	160	
	Repaired, but first stitch more than 1/4 inch from outsole junction	161	
	Repaired with one stitch in lieu of two or more stitches		264
Box toe	Misshapen, collapsed		265
Counters	Rolled or curled counter	162	
	Misshapen or collapsed	163	
Eyelets/speed laces	Number of eyelets/speed lacer not as specified	164	
	Not the same number of eyelets/speed laces in each row		
	Seriously affecting serviceability	165	
	Not seriously affecting serviceability		266
	Eyelets/speed lacers not properly spaced within the row	166	
	Misalignment between the rows to an extent interfering with proper lacing	167	
	Any eyelet/speed lacer not securely clinched	168	
	Any eyelet/speed lace malformed or damaged	169	
Removable cushion inserts	Missing, wrong size or width		267
Perforated holes	Components omitted		
	Seriously affecting serviceability	170	
	Not seriously affecting serviceability		268
	Incorrect number of holes		
	Seriously affecting serviceability	171	
	Not seriously affecting serviceability		269
	Incorrect placement of holes		270
	Incorrect number of rows and columns of holes		271
	Mesh or laminate not secured with adhesive or stitching	172	

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TABLE X. Defects - Continued.

Examine	Defect	Examine	
		Major	Minor
Pull tabs	Components omitted	173	
	Incorrect size	174	
	Incorrect location		
	Seriously affecting serviceability	175	
	Not seriously affecting serviceability		272
Comfort notch placement	Incorrect placement		
	Seriously affecting serviceability	176	
	Not seriously affecting serviceability		273

4.4.3 Component testing. The components shall be tested for the characteristics listed in table XI. The methods of testing specified where applicable shall be used. Unless otherwise specified in the contract or solicitation (see 6.2), the sampling plan shall be as follows:

Lot size (yards or units)	Sample size
800 or less	2
801 through 22,000	5
22,001 and above	8

TABLE XI. Component tests.

Component	Characteristic	Requirement Paragraph	Test Method
Upper leather	Physical characteristics	3.5.1	MIL-PRF-3122
	Shade evaluation		[1/]
	Softness		ISO 17235 [2/]
Insole	Weight	3.5.2 and table I	ASTM D3776/D3776M, option C
	Thickness		ASTM D1777 [3/]
	Tensile strength		ASTM D5034
	Bursting strength		ASTM D6797
	Elongation, percent (min)		ASTM D5034
Box toe and heel counters	Initial collapsing load	3.5.3, table II, and 3.5.4	SATRA TM83
	Resilience		SATRA TM83 [4/]
	Moisture resistance		SATRA TM83 [5/]
	Area shape retention		SATRA TM83
	Thickness		SATRA TM1

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TABLE XI. Component tests - Continued.

Component	Characteristic	Requirement Paragraph	Test Method
Nylon duck fabric	Physical characteristics Shade evaluation	3.5.5.1	MIL-DTL-32439 1/
Gusset fabric	Shade evaluation Weight Thickness Wales Courses Tear strength Breaking strength Elongation, percent	3.5.5.2 and table III	1/ ASTM D3776/D3776M, option C ASTM D1777 3/ ASTM D3787 ASTM D3787 ASTM D1424 ASTM D5034 ASTM D4964
Fabric liner and vamp lining	Martindale abrasion Dry bursting strength	3.5.5.3, 3.5.5.5 and table IV	ASTM D4966 6/ ASTM D6797
Back seam tape	As specified in A-A-52071	3.5.5.4	As specified in A-A-52071
Binding tape	Width Shade evaluation	3.5.5.6	Visual 7/ 1/
Backstay, ankle reinforcement tape and pull tab	Width Pull tab circumference Shade evaluation	3.5.5.7 3.5.5.8 3.5.5.7	Visual 7/ Visual 7/ 1/
Collar foam	Thickness Density	3.5.5.9 3.5.5.9	ASTM D1777 3/ ASTM D3574, test A
Shanks	Width Length	3.5.6 and table V	Visual 7/
Speed lace assembly	Shade evaluation	3.5.7	1/
Eyelets	Shade evaluation	3.5.7	1/
Laces	Length Properties as specified in A-A-55093	3.6	Visual 7/ As specified in A-A-55093
Outsole	Sole design/pattern Shade evaluation	3.7	Visual 1/

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TABLE XI. Component tests - Continued.

Component	Characteristic	Requirement Paragraph	Test Method
Outsole rubber compound	Abrasion	3.7.3 and table VII	ASTM D5963
	Hardness (shore A)		ASTM D2240 8/
	Stitch tear (dry)		ASTM D4786
	Stitch tear (Fuel B)		ASTM D4786 9/
	Die C tear strength		ASTM D624
	Ross flex, IRM 903		ASTM D1052
	Volume swell, Fuel A at 24 hours		ASTM D471
	Non-marking		SATRA TM223
	Tensile		ASTM D412
	Elongation		ASTM D412
Midsole	Density	3.7.4	ASTM D3574
	Hardness (shore A)		ASTM D2240
	Tensile strength		ASTM D3574
	Elongation, percent		ASTM D3574
	Tear strength, Die T		ASTM D3574
	Volume swell, w/IRM 903 (at 24 hours)		ASTM D471
	Compression set		SATRA TM64
	Ross flex at cycles (1 inch x 6 inches) at -5 °C		ASTM D1052
Thread, nylon	Shade evaluation	3.8	1/

FOOTNOTES:

1/ The color of the component shall be Coyote 498 when compared to the standard sample (see 3.2) when tested in accordance with AATCC EP9, option C as described in 4.6 of this specification.

2/ Test conducted with a 35-millimeter aperture.

3/ A thickness gauge of the dead-weight type equipped with a dial graduated to read directly to 0.001 inch shall be used. The presser foot shall be circular with a diameter of 1.129±0.001 inches and with moving parts weighted to apply a total load of 0.60±0.03 pound per square inch (psi) to the specimen. The anvil shall be not less than 1.129 inches in diameter. The presser foot and anvil surface shall be paned to within 0.001 inch and shall be parallel to each other to within 0.001 inch.

4/ Resilience is percentage retention of initial collapsing load after ten collapses.

5/ Moisture resistance is percentage retention of initial dry collapsing load after 1-hour immersion.

6/ Martindale tester; pressure applied: 12 kilopascals; visually examine specimens after each 3200 cycles; abradant changed after each 3200 cycles.

7/ Measured with a calibrated ruler/caliper.

8/ Thickness of test slab shall be as stated in the respective ASTM method.

9/ Test conducted after 46 hours in Fuel B (see ASTM D471) at room temperature. All testing for conformance purposes (see 6.2) shall use test slabs.

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4.4.4 End item testing. The finished boot shall be tested for the characteristics listed in table XII. The sample unit for all tests shall be one boot. All test results shall contain the individual values utilized in expressing the final result. For all tests, the sample size shall be five boots regardless of lot size and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

TABLE XII. End item tests.

Characteristic	Requirement Paragraph	Test Method
Boot height	3.11	Visual 1/
Weight	3.12	2/
Dry time	3.18	4.5.1

FOOTNOTES:

1/ Measured with a calibrated ruler/caliper

2/ Weighed on a calibrated scale

4.5 Methods of test.

4.5.1 Dry time testing. Boots shall be tested side-by-side with samples of the current boot (see 6.7.1) to allow for a comparison under similar environmental conditions. The dry time for the boots shall be conducted according to the following procedure:

a. All boots shall be laced and any packaging material removed before conditioning and measurements.

b. Dry mass measurements:

(1) All boots shall be conditioned at 70±5 °F and 65±3 percent RH, with a windspeed of 0.9 miles per hour, for a minimum of 4 hours.

(2) Load cell dry mass of each conditioned boot undergoing drying evaluation shall be taken and recorded on the load cell upon which drying will occur.

c. Boot soak:

(1) Each boot shall be submerged in a 5-gallon bucket filled with distilled water to cover the submerged boot for 30±5 minutes. Dive weights shall be inserted into the boots to keep each boot submerged for the soak period

(2) After the soak period, the boot shall be pulled from the water, the dive weight shall be removed, and the boot shall be held upside down with the toe higher than the heel to promote full drainage for 5 minutes.

(3) After draining (see 4.5.1.c.2), each boot shall be individually weighed and the mass recorded as the maximum mass, and the difference between the maximum mass and the dry mass shall be recorded as the water gained mass.

(4) Each boot shall be bagged to be taken to environmental test chamber.

d. Drying:

(1) Conditions in the chamber shall be 85±3 °F and 75±3 percent RH.

(2) Load cells shall be tared (zeroed) and the logging software shall be started to record mass and time.

(3) The chamber shall be opened, the boot shall be loaded on the associated load cell, and the chamber shall then be closed; repeat with other boots for evaluation (up to 3 boots at a time). All boots shall be loaded on the test cell in the same manner and orientation to reduce variability in results.

(4) Boots shall be allowed to dry for 6±0.5 hours.

(5) Data (dry mass and dry time) shall be recorded for analysis.

e. Repeat with remaining boots.

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4.6 Visual shade matching. The color and appearance of the boot shall match the standard sample when viewed using the AATCC EP 9, option C (see 6.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D_{65}, with a color temperature of 6500±200 Kelvin (K) illumination of 100±20 foot candles. Alternatively, the color and appearance of the boot shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D_{75}, with a color temperature of 7500±200K illumination of 100±20 foot candles in lieu of CIE Illuminant D_{65} (see 6.5.1). The boot shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856±200K.

4.7 Toxicity test. When the toxicity requirement (see 3.17) can be demonstrated with historical use data, toxicity testing may not be required on the finishing treatments used. If dermal toxicity testing is required (see 6.2), it shall be conducted in accordance with 40 CFR §798.2250 and 40 CFR §798.4100, which are consistent with OECD Guidelines for the Testing of Chemicals, Section 4, Test Nos. 404 and 406 (see 6.8).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boots are intended for wear by military personnel of the Department of Defense in hot weather (wet and dry) environments, garrison environments, and during combat.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, size, and width required (see 1.2).

c. The specific issue of individual documents referenced (see 2.2 and 2.3).

d. When first article is required (see 3.1.1, 4.2, and 6.3).

e. First article and conformance inspection AQLs (see 3.1.2, 4.2, and 4.3).

f. Applicable NSN and Universal Product Code.

g. Toxicity testing requirements (see 3.17).

h. Quality assurance standards (see 3.19).

i. Inclusion of specific instructions regarding arrangement for examinations, quantity, testing and approval (see 4.1.1).

j. First article samples required (see 4.2).

k. Component testing sampling (see 4.4.3).

l. Testing for conformance purposes (see table XI, footnote 9).

m. When toxicity testing is required (see 4.7).

n. Packaging requirements (see 5.1).

o. Shade criticality information (see 6.5.2).

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6.3 First article. When a first article inspection is required (see 3.1.1), it will be inspected and approved under the appropriate provisions of Federal Acquisition Regulation 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2017, option A of AATCC EP9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be, “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.”

6.5.1 Use of CIE illuminant D_{65} light source. In 2022, all services of the United States military accepted CIE Illuminant D_{65} as the new preferred light source for visual shade matching due to the supply chain issues with CIE Illuminant D_{75}.

6.5.2 Shade criticality. Some items may be deemed “non-shade-critical” by the contracting agency and alternative shade standards or information regarding shade may be specified in the contract or order (see 6.2). It is recommended that manufacturers refer to the contract or order to determine the criticality of the shade matching or alternate shade standards that are acceptable. Non-shade critical items are permitted to merely “approximate” the specified color, meaning that a common observer, not trained in color assessment but with average color vision under daylight conditions would accept the item as being the specified shade when the specimen and standard are held apart from each other and viewed at the same time.

6.6 Last sizes. Last sizes 3 through 16 cover the boot size range of 3 through 16 in this specification.

6.7 Known sources. Items that have been demonstrated to meet the requirements of the boots included in this specification are cited below. This citation is not intended to exclude any vendor, other item, or product that may meet the requirements of this specification.

Danner
www.danner.com
Boot model: Reckoning USMC Hot EGA

Rocky
www.rockyboots.com
Boot model: USMC Tropical Puncture Resistant Boot (RKC091)

6.7.1 Current combat boot. For the purposes of testing, the current combat boot to be used for dry time comparison is the Danner MEB.

6.7.2 Known source for sole design. The Vibram Panama II sole design has been demonstrated to meet the requirements of this specification, but alternate designs may be considered.

6.8 Toxicity testing. OECD Test No. 406, Skin Sensitization and OECD Test No. 404, Acute Dermal Irritation/Corrosion have been suggested to adequately assess the requirement in 3.17, but other methods may be used.

6.9 Subject term (key word) listing.

Cup sole

Direct attach

Footwear

Insole

Leather

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Nubuck

Outsole

Rubber

Shank

CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Review activities:
Army – AV, MI
Navy – AS, CG1, NU
Air Force – 03, 70
DLA – CT

Civil agency:
GSA – FAS

Preparing activity:
Navy – MC
(Project 8430-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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